Assessing the Robustness of CD-Type Tests During Stress Episodes

ETC5860

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1 Introduction

1.1 Topic and Importance of the Project

In panel data, cross-sectional entities are often not entirely independent of one another and may exhibit non-linear forms of dependence. Hence, modelling for cross-sectional dependence is useful to properly characterize the *strength* and *nature* of the interactions present within a panel. Existing approaches include spatial models, fixed effects, and multifactor models. However, because the true dependence structure is typically unknown, it is unlikely that such models can fully capture all sources of cross-sectional dependence (Pesaran & Xie, 2026). In particular, these models are typically designed to capture linear dependence structures and may therefore fail to account for non-linear forms of dependence commonly observed in financial and macroeconomic data, especially through volatility dynamics or higher-order moments. One possible approach is to model these non-linear dependence structures directly. However, doing so typically requires additional assumptions regarding the underlying dependence mechanism and may be difficult to estimate reliably in high-dimensional panel settings.

As a result, non-linear forms of dependence may remain in panel residuals even after selecting models intended to account for cross-sectional dependence, posing challenges for residual-based diagnostic tests of cross-sectional independence, which are typically constructed to detect linear cross-sectional correlation. Nevertheless, such diagnostic procedures remain essential for assessing whether cross-dependence structures have been adequately accounted for and whether reliable inference can be conducted (Pesaran & Xie, 2026). Recent work by Dalla et al. (2022) and Giraitis et al. (2024) addresses this issue by developing robust correlation-based tests that remain asymptotically valid under broader forms of heterogeneity, including non-linear and heteroskedastic dependence structures. However, these results have thus far been developed only in univariate and bivariate time series settings, thereby providing an avenue for extension to panel data.

2 Background and Motivation

2.1 Preliminary Literature Review

Early approaches to testing cross-sectional dependence include likelihood based methods such as the Breusch-Pagan Lagrange multiplier (LM) test, which is based on the average of squared pairwise residual correlations. While valid when the cross-sectional dimension N is fixed and the time dimension $T \to \infty$, these tests suffer from substantial size distortions when N becomes large (Pesaran, 2004, 2021). Pesaran (2004) showed that re-scaling and re-centering allowed the LM test to remain valid when both $T \to \infty$ and $N \to \infty$, although the resulting statistic still suffers substantial size distortions when T is fixed and $N \to \infty$.

As a result, Pesaran (2004) proposed the CD statistic, which has since become a popular alternative for testing the null hypothesis of cross-sectional independence. Unlike the LM and scaled-LM statistics, the CD statistic is based on a scaled sum of all pairwise cross-correlation coefficients of OLS residuals, and under symmetrically distributed errors, remains correctly sized even when T is small and N is large, or when $N, T \to \infty$ in any order (Pesaran, 2004). In settings where cross-sectional dependence is local in nature, substantially greater power can be achieved using the spatial extension of the test, denoted CD(p), which restricts the summation to pairwise cross-correlations between neighbouring units up to order p (Pesaran, 2004). Importantly,

both the CD and CD(p) statistics are robust to time-varying unconditional heteroskedasticity, provided that the unconditional means of the panel variables remain constant over time, and are well-sized in finite samples even under substantial departures from symmetrical errors required for its finite-sample validity (Pesaran, 2004, 2021).

More recent work by Juodis and Reese (2022) demonstrated that the CD statistic is invalid and results in over-rejection of the null of cross-sectional independence when applied to residuals from models with multifactor error structures. In particular, they showed that cross-dependence induced by latent factors or heteroskedastic errors introduces the Incidental Parameters Problem (IPP), whereby estimation of a large number of nuisance parameters, such as time-fixed effects or latent factors, affects the asymptotic distribution of the test statistic. To address this issue, they proposed weighted and screened versions of the CD statistic, denoted CD_W and CD_{W+}, that mitigate the incidental parameter bias by randomizing the pairwise correlation coefficients and selectively emphasizing sufficiently large correlations to improve power under cross-sectional dependence. Subsequently, Pesaran and Xie (2026) provided an analytical approach to this issue by proposing a bias-corrected CD^* test, where they instead conditioned on the estimated unobserved factors and their loadings.

2.2 Motivation and Aim of This Project

A major limitation of the CD-type tests and correlation-based diagnostics more generally is their reliance on the assumption that the underlying errors are independently and identically distributed (i.i.d.). When this assumption is violated, as highlighted by Anderson (1979), Dalla et al. (2022), and Taylor (1984), such tests may exhibit substantial size distortions and no longer provide reliable evidence regarding the absence of cross-correlation or serial correlation, particularly when uncorrelated data exhibits non-stationarity or non-linear dependence.

Concerns regarding the validity of standard correlation tests under non-i.i.d. data have long been recognized in the time series literature. Taylor (1984) and Cumby and Huizinga (1992) developed robust procedures for testing zero serial correlation in the presence of heteroskedasticity and other departures from the i.i.d. assumption, based on the t-type statistic and corresponding portmanteau (Q-type) statistics of Box and Pierce (1970). Haugh (1976) and Haugh and Box (1977) extended these ideas to the bivariate case by proposing modified t-type and Q-type statistics for testing zero *cross-correlation* using sample cross-correlations. Kyriazidou (1998) later developed corrected versions of these cross-correlation procedures that extend the univariate results of Cumby and Huizinga (1992) to the bivariate setting. However, their proposed test still relies on restrictive assumptions, such as stationarity, and excludes unconditional heteroskedasticity even though constant unconditional variance in the error term is often inconsistent with observed financial and macroeconomic data (Dalla et al., 2022; Giraitis et al., 2024). More recent work by Dalla et al. (2022) relaxes many of these assumptions by developing robust t-type and Q-type tests for zero serial correlation or cross-correlation that remain asymptotically valid under heteroskedastic and dependent processes, provided that heteroskedasticity evolves smoothly and deterministically. Subsequently, Giraitis et al. (2024) extended these robust procedures to a broader class of uncorrelated processes exhibiting general heterogeneity, allowing cases for heteroskedasticity to be non-smooth and stochastic.

Despite these advances, such robust procedures have largely remained confined to univariate and bivariate time series settings, with little work extending these ideas to panel data diagnostics for cross-sectional dependence. The goal of this project is, therefore, to adapt the results of Dalla et al. (2022) and Giraitis et al. (2024) to the widely used CD statistic of Pesaran (2004, 2021)

for testing cross-sectional dependence in panel data models with a multifactor error structure to produce tests that are robust under conditional heteroskedasticity and general heterogeneity, thereby enhancing their applicability in empirical work. The insights obtained from this analysis can then be further extended to other correlation-based statistics in the panel data literature, such as the scaled-LM, CD(p), CD$_{W+}$, and bias-corrected CD* statistics described above, among others.

2.3 Contribution

Our contribution to the literature will be summarized as follows:

1. Simulation studies assessing the finite-sample performance of CD-type statistics under conditional heteroskedasticity and general heterogeneity in correlated and uncorrelated panels with multifactor error structures.
2. Robust corrections to these CD-type statistics following the framework of Dalla et al. (2022) and Giraitis et al. (2024) in the event of test over-rejection in uncorrelated panels (size distortions).
3. Application of the robustified statistics to real house price changes across Metropolitan Statistical Areas in the United States with explicit consideration of a time period that includes major financial and economic dislocations.

3 Theoretical framework of the Project

3.1 Preliminary definitions and theorems

The proposed statistics are based on the framework of the standard CD statistic of Pesaran (2004, 2021) which is based on the following panel data model

$$y_{it} = \alpha_i + \boldsymbol{\beta}_i' \mathbf{x}_{it} + u_{it}, \quad \text{for } i = 1, \ldots, N,\ t = 1, \ldots, T,$$

where i and t denote the cross-sectional and time dimensions of the panel, respectively. The parameters α_i are individual-specific intercepts, $\mathbf{x}_{it}$ is a $k \times 1$ vector of observed regressors, and $\boldsymbol{\beta}_i$ allows for slope heterogeneity across units. Cross-sectional dependence in the error term, u_{it}, may arise from a variety of sources, including spatial dependence and omitted unobserved common factors, conditional heteroskedasticity, among others. The regressors may include lags of y_{it} and may be stationary or exhibit unit roots. The model reduces to the standard fixed effects model when slope coefficients are homogeneous across units, i.e., $\boldsymbol{\beta}_i = \boldsymbol{\beta}$.

The CD statistic used to test cross-sectional dependence — or more precisely, cross-*correlation* — is based on a scaled sum of all pairwise cross-sectional correlation coefficients computed from the OLS residuals $e_{it} = y_{it} - \hat{\alpha}_i - \hat{\boldsymbol{\beta}}_i' \mathbf{x}_{it}$ and is given by

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}$$

where

$$\hat{\rho}_{ij} = \hat{\rho}_{ji} = \frac{\sum_{t=1}^{T} e_{it} e_{jt}}{\left(\sum_{t=1}^{T} e_{it}^2\right)^{1/2} \left(\sum_{t=1}^{T} e_{jt}^2\right)^{1/2}} \tag{1}$$

Pesaran (2004, 2021) showed that the following assumptions are sufficient to establish the finite and asymptotic validity of the CD statistic for stationary $\mathbf{x}_{it} \sim I(0)$:

Assumption 1. For all i, the errors, u_{it}, are serially independent with $E(u_{it}) = 0$ and $\text{Var}(u_{it}) = \sigma_i^2$, where $0 < \sigma_i^2 < \infty$.
Assumption 2: Under the null hypothesis given by

$$H_0 : \ u_{it} = \sigma_i \varepsilon_{it}, \quad \text{where } \varepsilon_{it} \sim i.i.d.(0,1) \text{ for all } i,t,$$

the errors, ε_{it}, are symmetrically distributed around 0.
Assumption 3. The regressors, $\mathbf{x}_{it}$, are strictly exogenous, such that

$$E(u_{it} \mid \mathbf{X}_i) = \mathbf{0}, \quad \text{for all } i,t,$$

where $\mathbf{X}_i = (\mathbf{x}_{i1}, \ldots, \mathbf{x}_{iT})'$ and $\mathbf{X}_i'\mathbf{X}_i$ is positive definite.
Assumption 4. $T > k+1$, and the OLS residuals, e_{it}, are not identically zero.

Theorem 1 Under assumptions 1-4, $E(\hat{\rho}_{ij}) = 0$ and $E(CD) = 0$ for all N and $T > k+1$.
Theorem 2 Under assumptions 1-4, $CD \xrightarrow{d} N(0,1)$ as $N, T \to \infty$ in any order.

Our main focus is to relax Assumption 2 by introducing non-linear cross-dependence in the errors, u_{it}, through conditional heteroskedasticity (e.g. ARCH/GARCH-type dynamics) that latent factors, fixed effects, and spatial structures fail to capture. We then consider the theoretical finite and asymptotic properties of the CD statistic under these variance structures and conjecture that the validity results established in Theorems 1 and 2 may no longer hold under such structures.

To illustrate, suppose that the true error process is given by

$$\begin{aligned} u_{it} &= \sigma_{it}\varepsilon_{it}, \\ \sigma_{it} &= \lambda_i h_t, \\ h_t^2 &= \omega + \alpha \bar{u}_{t-1}^2 + \beta h_{t-1}^2, \quad \text{for all } i,t, \end{aligned}$$

where the errors $\varepsilon_{it} \sim i.i.d.(0,1)$ for all i, and cross-sectional dependence is induced through the conditional error variance σ_{it}, which evolves according a 'common' GARCH-type volatility process. Then

$$E(u_{it}u_{jt}) = 0$$

and

$$\begin{aligned} \text{Cov}(u_{it}^2, u_{jt}^2) &= E(u_{it}^2 u_{jt}^2) - E(u_{it}^2)E(u_{jt}^2) \\ &= \lambda_i^2\lambda_j^2 E(h_t^4) - \lambda_i^2\lambda_j^2[E(h_t^2)]^2 \\ &= \lambda_i^2\lambda_j^2 \,\text{Var}(h_t^2). \end{aligned}$$

Hence, the errors u_{it} are uncorrelated but dependent. Since

$$E(u_{it}u_{jt}) = 0,$$

the sample cross-correlation coefficients, $\hat{\rho}_{ij}$, in (1) remain correctly centered and the results of Theorem 1 continue to hold. However, because

$$E(u_{it}^2 u_{jt}^2) = \lambda_i^2\lambda_j^2 E(h_t^4),$$

the variance of the numerator of the sample cross-correlation coefficient, $\hat{\rho}_{ij}$, depends on the common volatility process, h_t. Consequently, the standard normalization underlying the limiting distribution in Theorem 2 may no longer hold under such non-linear dependence structures, resulting in size distortions in the standard CD test.

The same issue arises in testing for zero cross-correlation between pairs of uncorrelated but dependent time series $\{x_t, y_t\}$, where Dalla et al. (2022) and Giraitis et al. (2024) proposed the following robust self-normalized statistic

$$\tilde{t}_{xy,k} = \frac{\sum_{t=k+1}^{n}(x_t - \bar{x})(y_{t-k} - \bar{y})}{\left(\sum_{t=k+1}^{n}\left[(x_t - \bar{x})(y_{t-k} - \bar{y})\right]^2\right)^{1/2}}.$$

The specific second-moment normalization ensures that the statistic is asymptotically normal and correctly sized under conditional heteroskedasticity and more general forms of heterogeneity.

4 Proposed Simulation Study

Our assessment of the finite sample properties of CD-type tests is grounded on rejection rates, which is computed as the proportion of instances where the test rejects the null hypothesis, conditional on the degree of cross-sectional dependence. Under the null and alternative hypotheses, these correspond to the test's size and power, respectively. Ideally, rejection rates would be high for the assessed test in cases of strong cross-sectional dependence in the errors, and close to the nominal significance level in cases of no cross-sectional dependence. Precaution is taken in assessing the power of CD-type tests in light of the bias identified by Pesaran and Xie (2026) under strong latent factor dependence corresponding to large values of γ_i in (3). The robustness of the test will be assessed through its ability to maintain its nominal size in the presence of conditional heteroskedasticity and general heterogeneity for a given N and T. In the event where a test fails to maintain its nominal size, we consider robust modifications following the approach of Dalla et al. (2022) and Giraitis et al. (2024).

The assessment of the CD-type tests will be conducted using a large number of Monte Carlo experiments across different combinations of N and T. For each (N, T) pair, a fixed number of replications will be performed, with a suitable burn-in period included to reduce the influence of initial conditions. To ensure consistency in assessing the finite sample properties of the various correlation-based tests, the experiments will be based on the following data generating process (DGP), adapted from Pesaran (2021):

$$y_{it} = \mu_i(1 - \beta_i) + \beta_i y_{i,t-1} + u_{it}, \tag{2}$$

$$u_{it} = \gamma_i f_t + \sigma_{it}\varepsilon_{it}, \quad i = 1, 2, \ldots, N;\ t = 1, 2, \ldots, T, \tag{3}$$

where f_t denotes an unobserved common factor that induces cross-correlation across units, and γ_i its associated factor loading. The DGP is subsequently modified to introduce GARCH-type dynamics or heterogeneity effects in the error term, u_{it}, through the volatility component, σ_{it}. The size and power will then depend on the following null and alternative hypotheses:

$$H_0 : \gamma_i = 0, \qquad \text{for all } i,$$
$$H_1 : \gamma_i \neq 0, \qquad \text{for some } i,$$

with rejection occurring whenever

$$|CD\text{-}type_{stat}| > c_\alpha.$$

Hence, any size distortions that arise is because the nominal critical value c_α fails to deliver the rejection probability α in the presence of non-linear dependence.

4.1 Preliminary Simulation Results

Preliminary simulations were carried out to reproduce the size and power results reported in Pesaran (2021) for the standard CD test under different specifications of the DGP above. The objective is to obtain consistent results for cases where non-linear dependence is not present in the data generating process, ensuring consistency with the existing literature before introducing such dependencies. The results are then used to examine the failure of the CD statistic once non-linear dependence is included, and inform approaches to accommodate them in the new statistic.

Three cases from the paper are presented. Case 1: heterogeneous AR(1) with normal errors, Case 2: heterogeneous AR(1) with chi-squared errors, Case 3: time-varying unconditional heteroscedasticity. Both size and power are reported. Results show correct size and monotonically increasing power for the CD statistic in all three cases, which aligns with the results in the original paper.

Table 1: Size and Power of CD test under normal errors

T/N	5	10	20	30	50	100
Size $\gamma_i = 0$						
5	0.066	0.069	0.070	0.063	0.065	0.064
10	0.059	0.060	0.053	0.056	0.057	0.050
20	0.047	0.050	0.052	0.045	0.053	0.047
30	0.051	0.054	0.045	0.049	0.058	0.047
50	0.058	0.055	0.054	0.050	0.054	0.044
100	0.048	0.057	0.055	0.045	0.054	0.043
Power $\gamma_i \sim iid\, U(0.1, 0.3)$						
5	0.098	0.107	0.234	0.328	0.432	0.633
10	0.076	0.182	0.317	0.433	0.705	0.910
20	0.084	0.379	0.462	0.619	0.944	0.990
30	0.101	0.275	0.714	0.887	0.978	1.000
50	0.189	0.537	0.832	0.948	0.999	1.000
100	0.202	0.617	0.968	1.000	1.000	1.000

The DGP is specified as $y_{it} = \mu_i(1-\beta_i) + \beta_i y_{i,t-1} + u_{it}$, $u_{it} = \gamma_i f_t + \varepsilon_{it}$, where $\beta_i \sim iid\, U(0,1)$, $\mu_i = \varepsilon_{i0} + \eta_i$, $\eta_i \sim iid\, N(0,1)$, and $\varepsilon_{it} \sim iid\, N(0,1)$. The CD test statistic is computed as $\text{CD} = \sqrt{\frac{2T}{N(N-1)}} \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \hat{\rho}_{ij}$, where $\hat{\rho}_{ij}$ is the pairwise sample correlation coefficient between e_{it} and e_{jt} for all $i \neq j$. Test is conducted at the 5% 2-sided significance level under null of no error cross-sectional dependence ($\gamma_i = 0$). Each experiment (T, N) is carried out using 2000 replications with the first 200 time periods discarded as burn-in.

Table 2: Size and Power of CD test under $\chi^2(1)$ errors

T/N	5	10	20	30	50	100
Size $\gamma_i = 0$						
5	0.068	0.064	0.077	0.059	0.070	0.065
10	0.056	0.057	0.056	0.062	0.047	0.059
20	0.052	0.055	0.051	0.047	0.047	0.033
30	0.052	0.054	0.055	0.042	0.049	0.043
50	0.048	0.048	0.043	0.040	0.052	0.056
100	0.049	0.042	0.049	0.050	0.046	0.051
Power $\gamma_i \sim iid\, U(0.1, 0.3)$						
5	0.142	0.199	0.419	0.616	0.731	0.878
10	0.115	0.227	0.516	0.746	0.909	0.981
20	0.144	0.393	0.562	0.835	0.980	1.000
30	0.108	0.349	0.813	0.917	0.994	1.000
50	0.305	0.519	0.807	0.991	1.000	1.000
100	0.525	0.707	0.988	1.000	1.000	1.000

The DGP is the same as that specified in the footnote of Table 1, except that $\varepsilon_{it} \sim iid\, \chi^2_{(1)}$.

Table 3: Size and Power of CD test under time-varying unconditional heteroskedasticity

T/N	5	10	20	30	50	100
Size $\gamma_i = 0$						
5	0.093	0.064	0.066	0.061	0.065	0.065
10	0.060	0.053	0.054	0.056	0.047	0.058
20	0.031	0.044	0.050	0.050	0.046	0.051
30	0.045	0.042	0.047	0.047	0.053	0.047
50	0.046	0.051	0.048	0.052	0.052	0.051
100	0.042	0.045	0.061	0.047	0.053	0.060
Power $\gamma_i \sim iid\, U(0.1, 0.3)$						
5	0.094	0.115	0.155	0.356	0.458	0.694
10	0.079	0.139	0.320	0.400	0.491	0.859
20	0.083	0.191	0.351	0.521	0.755	0.963
30	0.086	0.169	0.367	0.654	0.802	0.988
50	0.058	0.135	0.395	0.662	0.947	1.000
100	0.090	0.210	0.704	0.924	0.999	1.000

The DGP is the same as that specified in the footnote of Table 1, except that $\beta_{it} \sim iid\, U(0, 1)$ and $\sigma_{it} \sim iid\, \sqrt{\chi^2(2)/2}$ for $t = 1, ..., T$.

5 Proposed Empirical Study

The assessed tests (before and after robustification) are then applied to real house price changes across Metropolitan Statistical Areas in the United States to detect evidence of conditional heteroskedasticity. We consider a time-period with major financial and economic dislocations where underlying volatility is high and possibly common among cross-sectional units. After accounting for common factors, spatial effects, serial correlation, and seasonalities, among others, we expect to find evidence of conditional heteroskedasticity in the data using these tests. In our analysis, we account for the tendency of CD-type tests to over-reject the null of cross-sectional independence in the presence of strong latent factors and asymmetric errors (Pesaran & Xie, 2026).

6 Timeline

Month	Task
June	Perform simulations
July	Apply to real-world data
August	Analyse results
September	Write thesis
October	Finalise thesis

7 References

Anderson, O. D. (1979). An introduction to bilinear time series models. *Journal of the Royal Statistical Society Series C: Applied Statistics*, *28*(3), 305–306. https://doi.org/10.2307/2347208

Box, G. E., & Pierce, D. A. (1970). Distribution of residual autocorrelations in autoregressive-integrated moving average time series models. *Journal of the American statistical Association*, *65*(332), 1509–1526.

Cumby, R. E., & Huizinga, J. (1992). Testing the autocorrelation structure of disturbances in ordinary least squares and instrumental variables regressions. *Econometrica*, *60*(1), 185. https://doi.org/10.2307/2951684

Dalla, V., Giraitis, L., & Phillips, P. C. B. (2022). Robust tests for white noise and cross-correlation. *Econometric Theory*, *38*(5), 913–941. https://doi.org/10.1017/S0266466620000341

Giraitis, L., Li, Y., & Phillips, P. C. B. (2024). Robust inference on correlation under general heterogeneity. *Journal of Econometrics*, *240*(1), 105691. https://doi.org/10.1016/j.jeconom.2024.105691

Haugh, L. D. (1976). Checking the independence of two covariance-stationary time series: A univariate residual cross-correlation approach. *Journal of the American Statistical Association*, *71*(354), 378–385. Retrieved May 6, 2026, from http://www.jstor.org/stable/2285318

Haugh, L. D., & Box, G. E. P. (1977). Identification of dynamic regression (distributed lag) models connecting two time series. *Journal of the American Statistical Association*, *72*(357), 121–130. Retrieved May 6, 2026, from http://www.jstor.org/stable/2286919

Juodis, A., & Reese, S. (2022). The incidental parameters problem in testing for remaining cross-section correlation. *Journal of Business & Economic Statistics*, *40*(3), 1191–1203. https://doi.org/10.1080/07350015.2021.1906687

Kyriazidou, E. (1998). Testing for serial correlation in multivariate regression models. *Journal of Econometrics*, *86*(2), 193–220. https://doi.org/10.1016/S0304-4076(97)00114-0

Pesaran, M. H. (2004). General diagnostic tests for cross section dependence in panels. *SSRN Electronic Journal*. https://doi.org/10.2139/ssrn.572504

Pesaran, M. H. (2021). General diagnostic tests for cross-sectional dependence in panels. *Empirical Economics*, *60*(1), 13–50. https://doi.org/10.1007/s00181-020-01875-7

Pesaran, M. H., & Xie, Y. (2026). How to detect network dependence in latent factor models? a bias-corrected cd test. *Econometric Theory*, 1–41. https://doi.org/10.1017/S0266466625100212

Taylor, S. J. (1984). Estimating the variances of autocorrelations calculated from financial time series. *Journal of the Royal Statistical Society Series C: Applied Statistics*, *33*(3), 300–308. https://doi.org/10.2307/2347707